Metamorphic Testing of a Multi-Agent Housing-Trends Application

> **Credits:** rewritten for the v2 results and cut to 10 pages by Paridhi Mittal, starting from Vidhi Jain's first report (commit 957224c). Lines by git blame: Paridhi Mittal 82%, Vidhi Jain 18%. Full attribution: CREDITS.md.

TSE Assignment — Final Report Market: Hyderabad / Secunderabad (the brief's optional Indian-metro route)

How to read the results. The headline results are the **contract-v2 sweeps** (`contracts/CONTRACT.md` v2.1.0-draft): a deterministic sweep (§7.1) and the same sweep with the local model switched on (§7.2), each over twelve relations, 72 base inputs and three seeds. Earlier sweeps, run before the two team implementations were merged, are kept with their diagnoses in `docs/DIAGNOSIS.md` and quoted here only where they explain a v2 result.

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1. Introduction

This report describes a six-agent application that reads a frozen snapshot of Hyderabad residential listings and reports how asking prices moved over twelve months, and the metamorphic suite built to find faults in it without knowing the correct answer to any query.

Given "3 BHK in Gachibowli, ₹80 lakh to ₹4 crore, the twelve months to August 2023", nobody can say what the correct median price per square foot was. Metamorphic testing replaces that missing oracle with relations between runs: reordering the input must not move the trend, duplicating every listing must double the counts and leave the medians alone, scaling prices by k must scale the medians by k. A violated relation exposes a fault without knowing any single right answer.

What the relations found.

- **Genuine agent faults, each fixed and re-measured:** geography widening that broke locality-set inclusion (M3, 9/36 → 0/36); a listing review that cut its batch before removing duplicates (I3 in LLM mode, 8/72 → 0/72); before the merge, an outlier threshold computed per *(worker, month)* (I3 and I4 on all 108 runs) and four prompts that leaked information a judgment must not depend on.
- **An unsound relation:** adding a constant to prices while a price band stays in place changes the selected population (`M1raw`, kept as the counter-example).
- **A limit of the architecture:** a model judge whose verdict changes which listings are counted cannot satisfy relations that change what it is shown (D1, D2, part of M1).
- **A threat to the headline:** the reported year-on-year move depends on the sufficiency threshold, and at the contract's threshold it has the opposite sign to the official index (§9).

2. Data source

Dataset	99acres.com residential listings, Hyderabad slice
Origin	Kaggle `arvanshul/gurgaon-real-estate-99acres-com`, titled *"Indian Real Estate – 99acres.com"*; despite the slug it covers Gurgaon, Hyderabad, Mumbai and Kolkata (verified on the dataset page)
Licence	Community Data License Agreement – Sharing – v1.0
Scrape date	≈ 26 August 2023 (latest `POSTING_DATE`)
Raw size	9,487 rows × 55 columns
Integrity	SHA-256 `9e81698322606c86`…, checked on every load

The raw file is never modified; `verify_snapshot()` aborts if the checksum changes. Months are calendar months in `Asia/Kolkata` (UTC would move 18 listings into the previous month). Full provenance: `data/SOURCE.md`.

Population. Sale listings of dwellings with a parseable `POSTING_DATE`:

```
raw rows                        9487
sale only (PREFERENCE=S)        7927   -1560
dwellings only                  5085   -2842
parseable POSTING_DATE (IST)    4975    -110
canonical listings              4975   (4574 with a usable price and area)
```

Listings without a usable price stay in the frame, flagged: they count in `listing_count` but not in the price statistics (§3).

Defects found and how they are handled (evidence in `contracts/Data.md`):

Defect	Evidence	Handling
`MIN_AREA_SQFT` holds **square metres**	`AREA` string ÷ column = **10.7639** at both the 5th and 95th percentile (3,531 rows)	area parsed from the display string
int32 sentinel and zero prices	`MIN_PRICE`/`MAX_PRICE` = 2,147,483,647 on 5 rows, 0 on 75	not price-valid
sale, rent and PG mixed	`PREFERENCE` S 7,927, R 1,467, P 93	sale only
`BEDROOM_NUM` missing on 29.7%	2,806 of the 2,814 are `Residential Land`	dwellings only
"Price on Request"	87 raw rows carry placeholder bounds	not price-valid
area-entry errors in the tail	a 27 sqft "villa" at ₹2.15 crore	**kept**, for cycle 2 to catch (§4.2)

The square-metre column is the consequential one: trusting the name would inflate every price-per-area figure 10.76×, and the ratio being identical at both tails is what separates mislabelling from noise.

The snapshot's own limit. It is a single scrape, so older months hold only listings still live on the scrape date: **96.4% of the window's listings fall in its final six months**. §9 shows this dominates the study's uncertainty. A **synthetic snapshot** with uniform monthly depth is built alongside; it is the only place with a genuine oracle.

3. The output contract

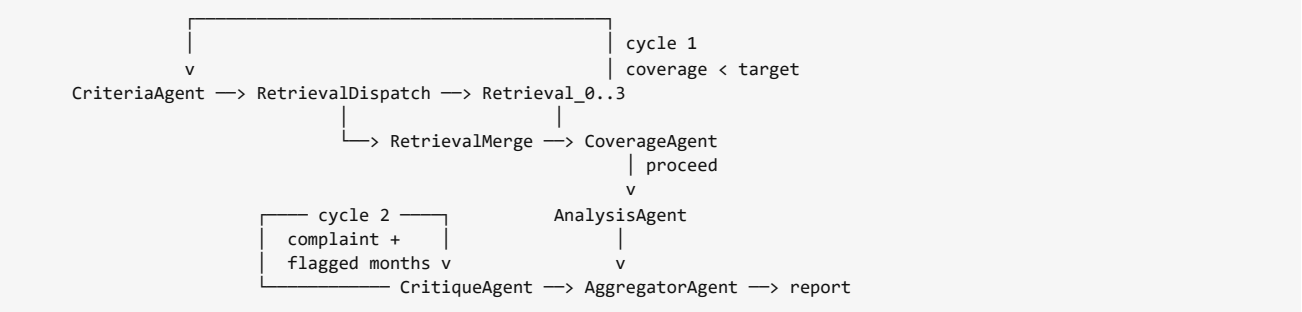
Fixed before any relation was written, because "the trend is unchanged" cannot be falsified until "the trend" is a named set of numbers (`src/contract.py`, `TrendReport`):

Field	Meaning
`window_months`	exactly twelve months, ascending
`monthly[]`	per month: `listing_count` (all matching listings), `price_valid_count`, `median_listing_price_inr` (median of (MIN + MAX) / 2), `median_price_per_sqft_inr`, `coverage_status` (ADEQUATE ≥ 30 usable listings / LIMITED / NO_DATA_IN_SNAPSHOT); extensions: min/max-price medians, ₹/sqft quartiles, `median_age_on_site_days`, `month_is_sufficient` (n ≥ 50), `is_partial_month`
`locality_rankings[]`	localities by median ₹/sqft, with explicit `rank`
`yoy_median_psf_pct`, `yoy_span_months`	move between the first and last **sufficient** month, and how many months that spans
`currency_unit`, `area_unit`	display units, carried so unit invariance is observable
`criteria`, `provenance`	the slice described; seeds, loop counts, verdicts, LLM accounting

Three choices exist so that relations mean something. **Medians, not means**: the price tail is contaminated, and a mean would fail every invariance relation for reasons unrelated to the agents. **Counts separate from prices**: duplication doubles counts and leaves medians fixed, and a fused statistic would hide exactly what the relations test. **The span of the "year-on-year" move is reported**: it runs between sufficient months, often far fewer than twelve. A missing median is `null`, never zero, and August 2023 is flagged partial. `median_age_on_site_days` is right-censored and length-biased, so it is not called days on market and no relation asserts on it.

4. Agent architecture

Six agents over a shared LangGraph state. The graph carries two cycles, both real conditional edges:



Agent	Reads	Writes
Criteria	request	`filter_spec`, `query_mode`
Retrieval ×4	`filter_spec`, `query_mode`, round	shard of matching listings
Coverage	merged listings	`coverage_score`, ACCEPT / RETRY / LIMITED, `widened_filter_spec`
Analysis	listings, `critique_complaint`	`trend_stats`
Critique	`trend_stats`	APPROVED / REJECTED / NEEDS_REVIEW, flagged months
Aggregator	`trend_stats`, coverage	`final_report`, narrative

4.1 The three required control-flow properties

(a) Feedback loop. Cycle 2, Critique → Analysis: a rejection names the months at fault, and Analysis recomputes only those, rejecting outliers by a robust z-score on log ₹/sqft. At most one recomputation; a problem still open after it is reported as NEEDS_REVIEW.

(b) Conditional, iterative step. Cycle 1, Coverage → Retrieval: `coverage_score` is the share of the twelve months with at least 30 usable listings. Below the target (0.5) Coverage widens one criterion — geography, then BHK, then price band, each at most once, at most two retries — and retrieval runs again; if the target cannot be reached it proceeds with **LIMITED** coverage and reports the gap. Geography widening keeps the locality constraint: each selected locality adds its own nearest neighbours, which preserves set inclusion (§8.4). Both cycles are capped, so the graph terminates.

(c) Genuine dependency. Criteria sets `query_mode`, and Retrieval runs a **different algorithm** for each: exact locality matching, or a haversine radius search that excludes listings without coordinates (5.3% of those with a usable price).

Several cooperating agents. Retrieval fans out to four parallel nodes, each scanning the shard given by a stable `sha256(seed:listing_id)`; a merge node reassembles them in canonical order. A test checks the answer is identical with 1, 2, 4 and 8 agents.

4.2 Why the loops are necessary rather than decorative

Cycle 1 is forced by the data. "3 BHK in Gachibowli" matches 120 listings, and only **two** months have 30 usable ones. Without the loop the system returns mostly empty months or silently answers a broader question. With it, coverage goes **0.167 → 0.250 → 0.333** as Gachibowli gains its 15 nearest localities and then BHK widens to 2–4 (120 → 432 → 696 listings), and the report states what was relaxed. A linear pipeline cannot do this: the decision to widen depends on an aggregate that exists only after retrieval. The run then stops at **LIMITED**: the whole snapshot could cover at most 7 months for this query, and the report says so instead of answering for the whole metro, which is what the pre-merge geography lever did.

Cycle 2 is forced by the area-entry errors. A 400 sqft villa at ₹8.3 crore is plausible alone and suspicious only against its month, so no blanket filter catches it, and validating every listing on every run would discard legitimate luxury stock. Escalating only for months whose aggregate looks wrong is why the complaint carries flagged months. On the query above, cycle 2 rejects **17 listings**, moves **4 monthly medians** (up to 11.6%) and changes the reported move from **−4.45% to −0.70%**; on the whole metro it rejects 132 and moves the headline from −6.41% to −7.74%.

4.3 Execution trace: both cycles firing

Abridged from `results/v2_deterministic/trace_example.txt` ("3 BHK in Gachibowli", seed 7):

```
 7. RetrievalMerge   agents=4  round=0  listings=120
 8. CoverageAgent    coverage=0.1667  covered=2 of 7 attainable  verdict=RETRY    + 15 neighbouring localities (cycle 1)
14. RetrievalMerge   agents=4  round=1  listings=432
15. CoverageAgent    coverage=0.25    covered=3 of 7             verdict=RETRY    BHK [3] -> [2, 3, 4] (cycle 1)
21. RetrievalMerge   agents=4  round=2  listings=696
22. CoverageAgent    coverage=0.3333  covered=4 of 7             verdict=LIMITED  decision=PROCEED
23. AnalysisAgent    mode=full  listings=696  move_pct=-4.45
24. CritiqueAgent    flagged 5 months  verdict=REJECTED -> recompute flagged months (cycle 2)
25. AnalysisAgent    mode=targeted recomputation  rejected=17  move_pct=-0.70
26. CritiqueAgent    flagged 2 months, none new  verdict=APPROVED
```

Cycle 1 fires twice and cycle 2 once, and the answer changes at step 25.

4.4 Where the local model sits

In LLM mode, agents delegate judgment to Qwen2.5-3B-Instruct (Q4_K_M) through Ollama: Analysis asks which **grey-zone** listings (robust |z| between 2.0 and 3.5) their own description contradicts; Critique asks which months look implausible; Aggregator writes the summary when there is a price change; Criteria can choose `query_mode` for a free-text request. **Every statistic stays deterministic Python**: the model supplies judgment and wording, never arithmetic. Temperature is 0 and the harness seed reaches the model on every call; a probe of 20 identical requests returned 20 identical replies, so an unchanged prompt means an unchanged judgment.

The team's design note is adopted in contract v2.1; its ten deliberate deviations — for example months from `POSTING_DATE` rather than `REGISTER_DATE__U`, which leaves 4 instead of 7 months coverable — are listed with evidence in `contracts/CONTRACT.md` §13.

4.5 The interface

The application is a Streamlit dashboard (`streamlit run app.py`) that runs the agent pipeline live on the frozen snapshot for whatever the user chooses:

- **Criteria panel:** city, BHK, localities, price band, window end, the retrieval algorithm (exact locality or a radius around a centre — the genuine dependency, made a user choice), and the agent controls (coverage target, retry and critique caps, number of retrieval agents, seed). Every request is validated against what the snapshot contains before an agent runs.
- **Trend tab:** the headline cards, the monthly median ₹/sqft and listing-count charts, the most expensive localities, and three views of the agents at work: *Show before critique* overlays the first analysis pass on the final one, shading the months cycle 2 changed (3 BHK Gachibowli: −4.45% → −0.70%); *How sure is the headline?* recomputes the yearly move at twelve sufficiency thresholds against the RESIDEX reference (§9); and *Watch the loops fire* steps through the run's trace beside a diagram that highlights the active agent and each cycle as it fires, with the answer as it stood at that step.
- **Relations tab:** the catalogue and every sweep's violation rates with their intervals; **Data tab:** the snapshot, its filter cascade and the defects.

Typography is Newsreader for headings with IBM Plex Sans and Plex Mono, in both light and dark themes. `tests/test_app.py` drives the dashboard headless, including all three agent views. An optional presentation dashboard over eight exported queries (`scripts/export_ui_data.py`, `results/ui/`) was also built; it shows saved results only, so the Streamlit app is the submitted application.

5. The metamorphic relation catalogue

Twelve relations across the three families (the minimum is eight), each a statement about `TrendReport` fields only. For every relation: the relation and why it should hold, the subdomain, the tolerance, and the fault a violation would expose. Floats are compared with `math.isclose` on unrounded values; counts exactly. Full definitions and the mapping to the design note's MR1–MR8: `contracts/Testing.md` §4.

5.1 Invariance

ID	Transformation	Relation, and why it should hold	Subdomain	Tolerance	Fault exposed
I1	permute the listing rows	every field unchanged: a trend is a property of the set of listings, not their order	all inputs	counts exact; floats 1e-9	order-dependent aggregation; unstable tie-breaks; shards leaking row order
I2	display ₹ → ₹ lakh, sqft → sq m	counts, statuses, rankings, % changes identical; ₹/sqft scales by the unit factor: units are presentation	all inputs; band kept in rupees	1e-9	hard-coded units; the square-metre confusion; rounding before percentages
I3	duplicate every listing	counts exactly double; per-month medians unchanged: repeating every value does not move a median	year-on-year fields excluded (doubling makes 25–49-listing months sufficient)	medians 1e-9; counts exact 2×	silent de-duplication; double counting; per-row judgments
I4	bijectively relabel listing IDs	every field unchanged: an identifier says nothing about price	all inputs	exact	logic keyed on IDs (shard hashing, prompts)
I5	display ₹ lakh → ₹ crore	as I2, factor 10^{-2}: the brief's Indian-metro case, "converting ₹ crore ↔ ₹ lakh … must not move the trend or the rankings"	lakh projection of each input	1e-9	hard-coded currency; rounding before percentages

5.2 Monotonic

ID	Transformation	Relation, and why it should hold	Subdomain	Tolerance	Fault exposed
M1	add ₹500,000 to both price bounds in the densest month	that month's price medians rise by ₹500,000; other months and all counts unchanged: a shift moves a median by the shift	**band-free projection**, so the shift cannot change which listings are selected	$\lvert\Delta - X\rvert \le$ 1e-6·X; others 1e-9	month-bucketing leaks; filters applied after the transform
M1raw	as M1, price band left in place	as M1 — **does not hold**: kept as the documented counter-example (§8.1)	none	as M1	none; the relation itself is unsound
M2	widen the price band 25% each side	each month's `listing_count` ≥ before: a superset filter cannot select fewer	inputs with a band	exact inequality	filter inversion; off-by-one on bounds; stale cached criteria
M3	keep the first half of the locality list	each month's `listing_count` ≤ before: a subset of places cannot hold more listings	inputs with ≥ 2 localities	exact inequality	leakage between shards; widening that breaks set inclusion (found, §8.4)
M4	halve the search radius	each month's `listing_count` ≤ before: a smaller disc is contained in the larger	radius inputs	exact inequality	wrong distance formula; unlocated rows leaking into a radius query

5.3 Directional

ID	Transformation	Relation, and why it should hold	Subdomain	Tolerance	Fault exposed
D1	shift the window forward one month	the eleven overlapping months' counts, statuses and medians match: a month's statistics cannot depend on which window contains it	both windows inside the data	counts exact; floats 1e-9	window off-by-one; bucketing drift; state carried between runs
D2	scale all prices (and the band) by k = 1.5	every price median scales by exactly k; % changes, rankings and counts unchanged: medians are scale-equivariant, and scaling the band keeps the population fixed	all inputs	1e-9	unit mishandling; non-linear aggregation

Stated in advance. M2, M3 and M4 change listing counts, which can change which cycle-1 levers fire and so the population for the whole window; they are also reported over the **control-path-stable subdomain**, where both runs took the same path. Cycle 2's relative outlier filter can also change the analysed population, so every run records how many listings it rejected in each run.

6. Statistical protocol

- **72 distinct base inputs**: 36 locality inputs (strided over window end, BHK, price band and locality set) and 36 radius inputs (centres at the six deepest localities, 3–5 km). Each relation runs on the inputs in its subdomain: at least 36 for every relation.
- **k = 3 fixed, recorded seeds `[11, 23, 47]`**, used for source and follow-up alike: **2,592 checks per sweep**, each two full pipeline runs.
- **The base input is the unit.** A base input counts as violated if any seed violates. Its three seeds are repeated measurements of one input, not independent samples, so intervals over runs would overstate precision (0/108 runs gives an upper bound of 0.034; 0/36 inputs, 0.096). The per-run rate and the number of inputs whose seeds disagreed are also recorded.
- **95% Wilson score intervals**, because rates sit at or near 0 and 1, where the normal approximation leaves [0, 1] or collapses to zero width. 0/72 gives **[0.000, 0.051]**.
- **Stopping rule.** A zero-violation relation is bounded at 0.051 (0.096 for the 36-input relations), enough to say it fails on fewer than one input in ten; every non-zero rate has an interval clearly separated from zero. The sweeps were therefore not extended.

7. Results

7.1 The deterministic sweep

`results/v2_deterministic/`, rebuilt by `python scripts/headline_table.py --v2` from the per-run records. Unit = base input.

MR	Family	Bases	Rate	95% CI	Stable-path rate
I1–I5	invariance	72	**0.000**	[0.000, 0.051]	0.000
M1	monotonic	72	**0.194**	[0.120, 0.300]	0.194
M2	monotonic	72	**0.000**	[0.000, 0.051]	0.000
M3	monotonic	36	**0.000**	[0.000, 0.096]	0.000
M4	monotonic	36	**0.000**	[0.000, 0.096]	0.000
M1raw	monotonic	71	**0.718**	[0.605, 0.810]	0.718
D1, D2	directional	72	**0.000**	[0.000, 0.051]	0.000

Seeds disagreed on no base input; the cycle-1 path diverged in 42 of 2,592 runs. M1raw has 71 inputs because one (`base10`) has no listing inside its window once the band is kept. Before M3's fix the same sweep gave M3 0.250 and M2 0.028 (`results/v2_before_m3fix/`); the sweep before the design-note alignment is `results/v2_before_note_alignment/`. On the **synthetic snapshot** M2 (0.028) and M3 (0.111) fail only on runs whose control path diverged, and are 0.000 on the stable subdomain.

7.2 The LLM sweep

`results/v2_llm/`: the same inputs, seeds and relations with the model agents on (`qwen2.5:3b-instruct-q4_K_M`, Ollama 0.40.4, weights digest `357c53fb…`, Kaggle T4). *Before fix* is `results/v2_llm_before_reviewfix/`, the same sweep before the I3 fault of §8.4 was fixed.

MR	Bases	Deterministic	Before fix	LLM	95% CI	Seeds disagree
I1, I2, I4, I5	72	0.000	0.000	**0.000**	[0.000, 0.051]	0
I3	72	0.000	0.111	**0.000**	[0.000, 0.051]	0
M1	72	0.194	0.375	**0.375**	[0.272, 0.490]	5
M2	72	0.000	0.000	**0.000**	[0.000, 0.051]	0
M3, M4	36	0.000	0.000	**0.000**	[0.000, 0.096]	0
M1raw	71	0.718	0.746	**0.746**	[0.634, 0.833]	1
D1	72	0.000	0.069	**0.069**	[0.030, 0.152]	0

MR	Bases	Deterministic	Before fix	LLM	95% CI	Seeds disagree
D2	72	0.000	0.278	**0.264**	[0.176, 0.376]	2

The cycle-1 path diverged in 43 runs (stable-path rates differ only for D1, 0.070, and D2, 0.268). Seeds now disagree on 8 (relation, input) pairs, because the seed reaches the model. The fallback rate — requests whose reply was unusable, so the deterministic rule was used — is **0.0383** over 17,164 model requests. Every violating M1, D1 and D2 run took the same control path in both runs, and in 128 of those 143 runs cycle 2 rejected a different number of listings: the model's listing review is where it changes the answer.

The sweep **replays without a GPU**: `python scripts/replay_llm_sweep.py` answers every model request from the committed reply cache, treats a missing reply as an error, and reproduces all 2,592 run records exactly.

	Deterministic	LLM
runs / base inputs	2,592 / 72	2,592 / 72
elapsed (s)	1,037 (Windows laptop)	1,596 (Kaggle T4)
coverage verdict	all 72 inputs LIMITED	—

8. Root-cause analysis

The three most-violated relations of the final system — the LLM sweep — are **M1raw (0.746)**, **M1 (0.375)** and **D2 (0.264)**. Each is attributed below to one of the brief's four causes; §8.4 lists the faults the other relations found. The full experiment log, including every pre-v2 diagnosis, is `docs/DIAGNOSIS.md`.

8.1 M1raw, 0.746 — an ill-conceived relation

Symptom. 53 of 71 inputs; typically `median_listing_price_inr moved 750,000.00, expected 500,000.00`, and in 24 runs a month's `listing_count` itself changed.

Evidence with no agents involved. Applying the criteria directly to the frame, before and after adding ₹500,000 to the shifted month:

Input	Band (₹ lakh)	Listings before → after	Median before	after	Moved
base00	40–150	2 → 1	14,377,250	14,500,000	**+122,750**
base01	80–400	15 → 16	18,000,000	16,877,250	**−1,122,750**
base09	40–150	15 → 14	8,253,000	8,051,500	**−201,500**
base05	50–200	11 → 11	14,000,000	14,500,000	+500,000

On `base01` the median **fell by ₹1.12 million** after every price in the month rose by ₹500,000. The band is applied after the shift, so listings cross its edges and the follow-up selects a different population; where no listing crosses (`base05`) the relation holds exactly. No agent takes part, so this is neither an agent fault nor a harness bug.

Control. D2 also transforms every price and holds at 0.000 deterministically, for one reason: it scales the band with the data.

Verdict: ill-conceived MR. Kept deliberately as the counter-example; M1, declared on the band-free projection, replaces it. Its rate barely moves between the deterministic (0.718) and LLM (0.746) sweeps, as an unsound relation's should.

8.2 M1, 0.375 — two causes, measured separately

Symptom. 27 of 72 inputs, e.g. `2023-03: median_listing_price_inr moved 750,000.00, expected 500,000.00`. The shift is applied to both price bounds, no band is in play, and every violating run took the same control path.

Cause 1 — the outlier filter is not shift-invariant (14 inputs, a data artifact). The deterministic sweep fails on 14 inputs, and the LLM sweep on all 14 of them. In every one of those violations cycle 2 rejected a different number of listings in the follow-up run; no input failed with the rejected count unchanged. Adding ₹500,000 to price changes ₹/sqft by a different amount per listing (₹1,000 at 500 sq ft, ₹100 at 5,000 sq ft), so the month's median and MAD of log ₹/sqft move and the relative threshold admits a different set. Two controls confirm it: with cycle 2 disabled M1 does not fail (0 of 15 runs pre-v2, the same inputs failing 15 of 15 with it on), and on the

synthetic snapshot, whose listings sit away from the threshold, M1 is 0.000. The relation is sound for the statistics; it fails through where real listings sit relative to a relative threshold. **Verdict: data artifact.**

Cause 2 — the model judges different prices (13 inputs). The other 13 inputs fail only with the model on; in 25 of their 31 violating runs the rejected count changed. The listing review shows the model each listing's price and ₹/sqft, so a shifted month is a different question and the model can answer it differently. This is D2's mechanism (§8.3). **Verdict: ill-conceived for an LLM-judged pipeline.**

8.3 D2, 0.264 — the model is shown different prices

Symptom. 19 of 72 inputs, e.g. `2023-03: median psf 10000.0 did not scale by 1.5 (got 14267.1)` — almost, but not exactly, ×1.5.

Evidence. D2 holds on all 72 inputs deterministically, so the statistics are scale-equivariant and the relation is sound for them. With the model, every violating run kept the same control path, and in 49 of 55 the number of rejected listings changed. Because the model is deterministic (§4.4), an unchanged prompt cannot produce a changed judgment; a prompt-sensitivity measurement on the pre-v2 system (`scripts/prompt_sensitivity.py`) found that D2 changes the review prompt on **every** input — more review prompts than any other relation — because every displayed price and ₹/sqft is multiplied by 1.5. Whether a description corroborates a price is a judgment *about* the price, so the model's verdict legitimately changes, a different set is rejected, and the median scales by k only up to that difference.

Prediction, then confirmation. Before the pre-v2 prompt fixes, D2 was predicted to stay violated because its prompt difference is irreducible, while the identity leak behind I4 was predicted to vanish. Afterwards I4 fell from 0.630 to 0.000 while D2 stayed violated, 0.556 → 0.593 (per run). D1 (0.069 in v2) has the same cause: shifting the window changes which months the review covers and the twelve-row table the critique reads.

Verdict: ill-conceived MR for this architecture. The relation holds for a pipeline whose selection does not depend on what an LLM judge is shown; this one deliberately lets the judge change which listings are counted. Stated generally: *a judge whose verdict changes the data selected creates a feedback path, and what it is shown decides which relations break.* We kept that design because invariance relations — renaming or duplicating a listing must not change the trend — admit no exception, while D1, D2 and M1 change data the judgment is legitimately about.

8.4 Other faults the relations found

Relation	Symptom	Cause, evidence	Category	Status
M3 (v2)	9/36: `2022-12: locality subset gave MORE listings (3 -> 4)`	geography widened around the *selection's* centre, so a subset and its superset widened around different points; on `base03` the subset gained five localities the superset did not	**genuine agent fault**	fixed: each locality widens around itself, skipping selected ones, which provably keeps widen(A) ⊆ widen(B); M3 0/36, M2 2/72 → 0/72
I3 (v2 LLM)	8/72: `price_valid_count 141 -> 283, expected exactly 2x`	the review cut each month's grey zone to its 12 highest-₹/sqft rows **before** removing duplicates; on `base23` the model saw 10 distinct listings in the source run and 8 after duplication, and a verdict reached only the twins inside the cut (283 = 2 × 141 + 1)	**genuine agent fault**	fixed and re-run on Kaggle: I3 0/72, D2 20 → 19, nothing else changed; it also closes the pre-v2 residual (0.176)
narrative (v2 LLM)	58 of 72 model summaries rejected	with no price change the model copied the facts block (49); the faithfulness guard rejected the radius it had been given (9)	model limitation; **genuine agent fault**	guard fixed; no model summary without a change. No relation compares the narrative
I3, I4 (pre-v2)	108/108 runs	outlier threshold computed per *(worker, month)*: 184 listings at 1–4 agents, 185 at 8	**genuine agent fault**	fixed; replicated on v2 (`results/v2_diagnosis/`)
I4, I3, I2, I1 (pre-v2 LLM)	I4 0.630, I3 0.417, I2 0.028	prompts carried listing IDs, display units, duplicated rows and an incomplete sort order	**genuine agent faults**	fixed by four prompt-hygiene rules

Relation	Symptom	Cause, evidence	Category	Status
M3 (pre-v2)	0.000	could not fail: geography widening sent every run to the whole metro	**harness/design blind spot**	became testable, then found the fault above
control-path report	M3's stable subdomain empty	signature compared human-readable notes containing criteria values	**harness bug**	fixed

All four of the brief's categories occur. Two of the findings concern our own instrumentation, which is the reason every relation's rate is reported with the rejected counts and control paths that explain it.

9. Threats to validity

The headline depends on the sufficiency threshold. The year-on-year move runs from the first sufficient month, early months hold 21–35 listings, and their medians are noise. Whole metro, statistics only (`scripts/threshold_sensitivity.py`):

n ≥	sufficient months	first	headline
10–25	12	2022-09	+87.18%
30	7	2022-10	+18.35%
40–60	**6**	**2023-03**	**−6.41%**
75–150	5	2023-04	+8.32%
200–300	4	2023-05	+3.87%

A 93.6-point spread, and the sign flips. **External cross-check:** NHB RESIDEX (*Press Release July–September 2023*, HPI at market price, based on asking prices) reports Hyderabad at **+6.8%**. At n ≥ 75 our snapshot gives +8.32%, the same sign and within 1.5 points; at the contract's n ≥ 50 it gives −6.41%, the **wrong sign**. We report −6.41% because that is what the contract fixed in advance — changing the threshold after seeing the index would fit the contract to the answer — and state that n ≥ 50 was a poor choice. It is one overlapping data point, not a series (`results/residex/cross_check.md`).

Other threats.

- **Single-point scrape, asking prices.** Older months hold only listings still live in August 2023 (survivorship), and the prices are asking prices.
- **Narrow queries end below the coverage target.** Keeping the locality constraint means no locality base input reaches 0.5 coverage, so many moves span two or three sufficient months; the report states it.
- **Relations stated over projections.** M1's band-free projection, I5's lakh projection and M2–M4's stable subdomain are reported with how often each excluded case occurs.
- **The LLM sweep ran on one machine.** It replays exactly from its cached replies, but a fresh run on other hardware may get slightly different replies even at temperature 0. The pre-v2 LLM sweeps used another population and per-run intervals; their rates are not compared with v2.
- **The model's role is limited by design.** It judges grey-zone listings and months and writes some summaries; 3.8% of its requests ended in the deterministic rule because the reply was unusable.
- **No PIN codes.** The dataset has no postal column; locality and radius are the geographic filters.
- **The synthetic snapshot is our own construction** and has no coordinates, so it cannot exercise radius search or geography widening.
- **The diagnosis rests on our own instrumentation**, which was wrong twice (§8.4); errors of the same kind would be equally hard for us to see.

10. Reflection

The brief asks which aspects an LLM could have produced without running the system.

Producible from the text alone: the shape of nearly everything. Extending the brief's eight example relations to twelve is routine. The architecture — planner, parallel retrievers, coverage check, analyser, critic, reporter, a feedback edge and a conditional retry — is a known pattern, and a model will justify why its loops are not decorative. The statistical protocol is standard. A model could also write a confident root-cause story for any results table, which is exactly the danger.

Requiring execution: every number, and every surprise. No reading reveals that `MIN_AREA_SQFT` holds square metres; that came from a ratio pinned at 10.7639 across both tails. Nor that our own cycles were once dead — the code compiled, the tests passed, and LangGraph silently discarded updates to undeclared state keys, visible only in a trace where the coverage agent said retry and nothing retried. The per-worker threshold showed up as 185 listings at eight workers against 184 at four; M1raw's flaw as a median *falling* ₹1,122,750 under a uniform increase; the review fault as 10 distinct listings becoming 8 when the data was duplicated; the threshold problem as a sign flip against an official index.

The sharpest version. We were wrong three times about why M1 failed, and every wrong answer was articulate. A relation passed 108 runs while the property it claimed did not hold. A fix that made M3 testable introduced a fault M3 then found. A model drafted every one of these changes, including the wrong ones; running the system is what decided which were right — which, in a project about testing without an oracle, is the subject matter. The use of AI throughout is recorded in `AI_USE.md`.

Appendix — reproducing this report

```
pip install -r requirements.txt
streamlit run app.py                                          # §4.5, the application
python scripts/headline_table.py --v2 --check                 # §7.1 from the run records, checked against this report
python scripts/headline_table.py --v2 --tag v2_llm --check    # §7.2 likewise
python scripts/replay_llm_sweep.py                            # §7.2 replayed offline, no GPU
python scripts/run_harness.py --tag v2_deterministic          # §7.1 from scratch, ~20-50 min
python scripts/run_harness.py --tag v2_synthetic --synthetic  # synthetic control
python scripts/run_harness.py --tag v2_llm --llm              # §7.2 from scratch: Ollama + the model
python scripts/threshold_sensitivity.py                       # §9 table
python -m pytest                                              # 157 tests
python scripts/verify_claims.py                               # re-derive the quoted figures
python scripts/run_checks.py --replay                         # all of the checks, logged to audit/test_logs/
```

Result sets: `results/v2_deterministic/`, `results/v2_llm/`, `results/v2_llm_before_reviewfix/`, `results/v2_before_m3fix/`, `results/v2_before_note_alignment/`, `results/v2_synthetic/`, `results/v2_diagnosis/`, `results/v2_sensitivity/`; pre-v2 sets and their diagnoses are described in `docs/DIAGNOSIS.md`. Shipped with them: the frozen snapshot and its checksum, the seeds `[11, 23, 47]`, and the model reply cache in `data/llm_cache/`. Contracts for every agent: `contracts/` (v2.1.0-draft). Every issue found, with how it was caught and fixed: `docs/ISSUE_LOG.md`. Each check run is logged in `audit/test_logs/`; the full set — tests, `verify_claims.py`, the three table checks and the offline LLM replay — also passed on a fresh clone with a new environment built from `requirements.txt`.